Worksheet — Hanging on Two Threads Simulation

Use with *Simulation - Hanging on two threads.html* · one mass on two threads tied to two fixed supports · equilibrium of three forces (Blocks 3–4)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/24-hanging-two-threads/** in a browser (offline: *Simulation - Hanging on two threads.html*). Choose an example at the top: 🧪 Lab · 🐦 Bird · 🚶 Walker · 🏮 Lantern · 🚦 Traffic · 🪧 Sign · 🧺 Washing. The **Example** list has more set-ups for each one. Set m, θ_L, θ_R and the thread with ▼ ▲ or type a value. Press ▶️ **Let go**: the mass hangs at rest, or a thread snaps.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 25 min) · **Part 3** homework (≈ 30 min). g = 9.81 N kg^{-1}. Answers to 3 s.f.

Angles θ_L and θ_R are measured **below the horizontal**. At rest: $T_L \cos\theta_L = T_R \cos\theta_R$ and $T_L \sin\theta_L + T_R \sin\theta_R = mg$.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 🧪 A 2.0 kg mass hangs from the middle of two threads, each at 30° below the horizontal. Is the tension in each thread half the weight, equal to the weight or more than the weight?			
2. 🧪 You make both threads flatter (smaller θ). Does the tension increase, decrease or stay the same?			
3. 🧪 Can the threads be pulled perfectly straight (θ = 0°) with the mass still hanging on them?			
4. 🐦 A 30 g bird lands on a cotton thread (breaks above 20 N) pulled almost straight: 0.4° each side. Does the thread hold?			
5. 🚦 A traffic light hangs from cables at 30° (left) and 60° (right). Which cable has the larger tension?			

6. 🚶 A tightrope walker stands near the left end: 12° on the left, 3° on the right. Which part of the wire has the larger tension — by a lot or a little?			

7. Calculate the tension in each thread when a 2.0 kg mass hangs at 10° on both sides. Check: 🧪 Lab → Example *2.0 kg, both threads at 10°*.

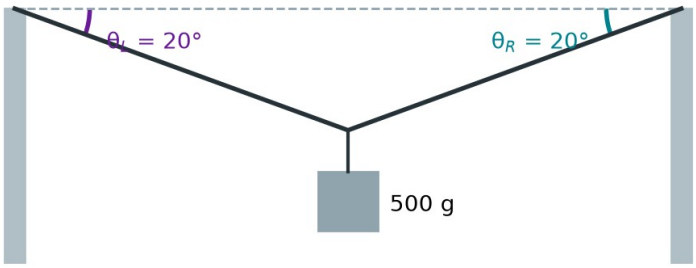
Working: FBD of the mass → $\Sigma F_y = 0$

Part 2 — During the lab: a string between two stands

Apparatus: two clamp stands with bosses (same height), strong string, two newton meters (0–10 N), 500 g mass hanger, protractor, metre rule.

Method: tie the string between the stands with a newton meter in each half; hang the 500 g mass from the middle. Move the stands apart or together to change the angle. Measure each angle below the horizontal with the protractor and read both newton meters.

⚠️ **Safety:** keep feet away from under the mass; clamp the stands to the bench (or hold the bases) — they are pulled towards each other. Do not go below about 10°: the readings rise very fast and can go off the newton-meter scale.



8. Calculate the tension at 20° — what each newton meter should read. Check: 🧪 Lab → *500 g on cotton thread, 20°*.

Working

Results (m = 0.500 kg, same angle on both sides). Theory: T = mg / (2 sin θ).

θ / °	left reading / N	right reading / N	mean T / N	T from theory / N	1 / sin θ
60					
45					

30					
20					
10					

9. Cotton thread breaks above 20 N. Calculate the smallest angle at which it can still hold the 500 g mass. Test it in the sim 1° above and 1° below.

Working: $2T \sin \theta = mg$

10. Set 20° on the left and 50° on the right (500 g). Calculate T_L and T_R, then compare with the newton meters and the sim (*500 g*, *angles 20° and 50°*).

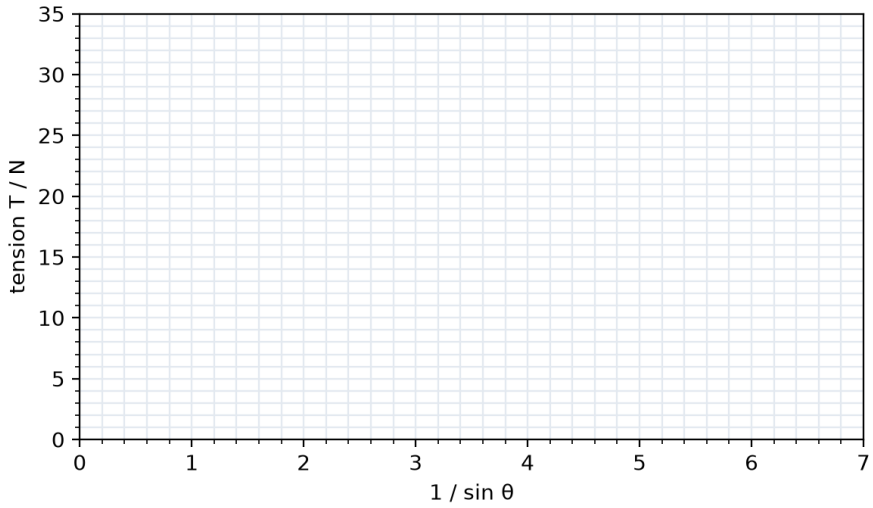
Working: $\Sigma F_x = 0$ and $\Sigma F_y = 0$

11. Explain why the newton-meter readings rise so much when the string is nearly straight.

Explanation

Part 3 — Homework: analyse

12. 🧪 Lab → *1.0 kg*, *unbreakable*. Set θ = 10°, 20°, 30°, 45°, 60° and 80° (same both sides), press ▶️ Let go and ● Record each one. Plot T against 1/sin θ on the grid, draw the best-fit line and find its gradient. Use it to find the mass (gradient = mg/2).



Gradient and mass

13. 🚶 A 70 kg tightrope walker makes the wire dip 5.0° on each side. Calculate the tension in the wire and how many times the walker's weight it is.

Working

14. 🚶 A nylon cord breaks above 2.0 kN. Calculate the smallest angle at which it could hold the 70 kg walker.

Working

15. 🚦 A 15 kg traffic light hangs from two cables at 30° (left) and 60° (right) below the horizontal. Calculate the tension in each cable.

Working: FBD → $\Sigma F_x = 0$ and $\Sigma F_y = 0$ (or a vector triangle)

16. 🐦 A 30 g bird sits in the middle of a wire that dips 3.0° on each side. Calculate the tension in the wire.

Working

17. 🏮 A 2.0 kg lantern hangs from the middle of a string; each half makes 40° with the horizontal. Calculate the tension.

Working

18. 🪧 (Problem set Q19) A 15 kg sign hangs from two cables, each at 40° to the horizontal. Calculate the tension in each cable.

Working

19. 🧺 3.0 kg of wet washing hangs in the middle of a line pulled tight (4° each side). Fishing line breaks above 100 N. (a) Show that the line snaps. (b) Calculate the smallest angle the line may make with the horizontal.

Working